1. Experiment in short

Number of nodes	6
Number of topologies	5
Location of the experiment	Werbellin Lake, Berlin, Germany (52.915103°N, 13.706159°E)
Measurement duration	600 seconds
Measurement granularity	1 second
Link quality measurements	On/Off status

2. Location

The experiment was conducted in June 2016, in the Werbellin lake, located in northern Germany. A Google map illustration of the experimental setup is shown in Fig. 1.

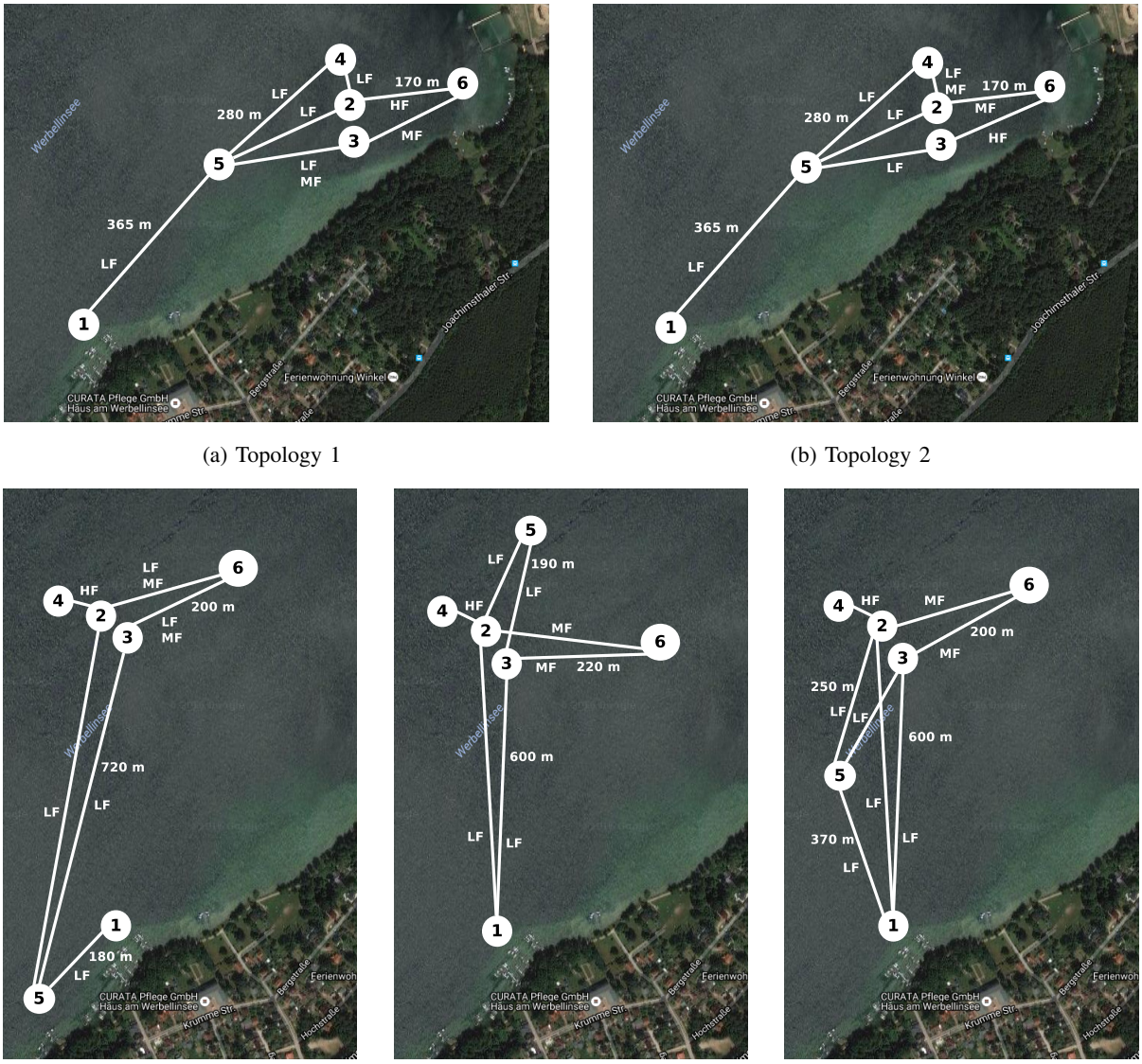
3. Topologies

TABLE I
Technologies available to each node in each topology and approximate deployment depth (between parentheses)

	Topology 1	**Topology 2**	**Topology 3**	**Topology 4**	**Topology 5**
Node 1	LF (10 m)	LF (10 m)	LF (10 m)	LF (10 m)	LF (5 m)
Node 2	LF, HF (10 m)	LF, MF (10 m)	LF, MF, HF (10 m)	LF, MF, HF (10 m)	LF, MF, HF (10 m)
Node 3	LF, MF (10 m)	LF, HF (10 m)	LF, MF (10 m)	LF, MF (10 m)	LF, MF (10 m)
Node 4	LF (10 m)	LF, MF (10 m)	HF (10 m)	HF (10 m)	HF (10 m)
Node 5	LF (5 m)	LF (10 m)	LF (5 m)	LF (10 m)	LF (10 m)
Node 6	MF, HF (3 m)	MF, HF (3 m)	LF, MF (3 m)	MF (3 m)	MF (3 m)

4. Description of the nodes

The experiment included 6 nodes, communicating using three different physical layer technology, incorporated by three different types of EvoLogics modems: the S2C 18-34 modem (low-frequency, LF); the S2C 48-78 modem (mid-frequency, MF); the S2CM HS model (high-frequency HF). A total of 10 modems was deployed: 5 LF, 3 MF and 2 HF. All modems employ EvoLogics's proprietary sweep-spread



(a) Topology 1

(b) Topology 2

(c) Topology 3

(d) Topology 4

(e) Topology 5

Fig. 1. Logical network topology configurations and locations of the nodes in the five scenarios considered the Werbellin Lake experiment. Each link is tagged with the technologies available to the connected nodes, which can be used to communicate over that link.

carrier modulation scheme. For further information, the interested reader is referred to EvoLogics's web site: https://www.evologics.de/en/products/acoustics/index.html The modems were integrated into the nodes in as shown in Fig. 1 and as indicated in Table I.

5. Tested protocols and schemes

The deployment was employed to test a multimodal routing algorithm aimed at distributing traffic over multimodal links in an optimal way, in order not to create bottlenecks in the network while at the same time allotting transmit opportunities fairly to nodes that do not have multiple technologies or routes available.

6. Link quality measurements

The network links are evaluated by means of on/off measurements, which indicate whether a packet transmitted at a given epoch would be received or not.

7. Available data

The measured data is provided in file `BerlinMultimodal-06-16.mat`, which includes all topologies presented in Fig. 1. For each topology, we provide 600 measurement epochs, given at constant intervals of 1 second starting from the beginning of each experiment. The file contains four types of matrices:

- the location matrices `LocMat1`...`LocMat5`, of size 600×6×3;
- the topology measurement matrices `TopMat1`...`TopMat5`, of size 600×6×6×3;
- the adjacency matrices `AdjMat1`...`AdjMat5` of size 600×6×6;
- the technology matrices `TechMat1`...`TechMat5` of size 600×6×3.

Additionally, for each matrix type, we concatenate all matrices into a single matrix, respectively named:

- `FullLocMat` of size 3000×6×3;
- `FullTopMat` of size 3000×6×6×3;
- `FullAdjMat` of size 3000×6×6;
- `FullTechMat` of size 3000×6×3.

Using Matlab/Octave matrix notation:

- `LocMat1(t,i,:)`...`LocMat5(t,i,:)` or `FullLocMat(t,i,:)` are the UTM coordinates (zone 33U) and the depth of node i at time instant t;
- `TopMat1(t,i,j,p)`...`TopMat5(t,i,j,p)` or `FullTopMat(t,i,j,p)` is 1 if the link $i \to j$ is available using technology p at time t, and 0 otherwise;
- `AdjMat1(t,i,j)`...`AdjMat5(t,i,j)` or `FullAdjMat(t,i,j)` is equal to 1 if there exists a topology link between nodes i and j at time t;
- `TechMat1(t,i,p)`...`TechMat5(t,i,p)` or `FullTech(t,i,p)` is equal to 1 if node i can use physical layer technology p at time t.

8. Reference to detailed description

Further details about this experiment are given in the following paper:

- R. Diamant, P. Casari, F. Campagnaro, O. Kebkal, V. Kebkal, M. Zorzi, "Fair and Throughput-Optimal Routing in Multimodal Underwater Networks," in *IEEE Transactions on Wireless Communications*, vol. 17, no. 3, pp. 1738–1754, March 2018. DOI: 10.1109/TWC.2017.2785223